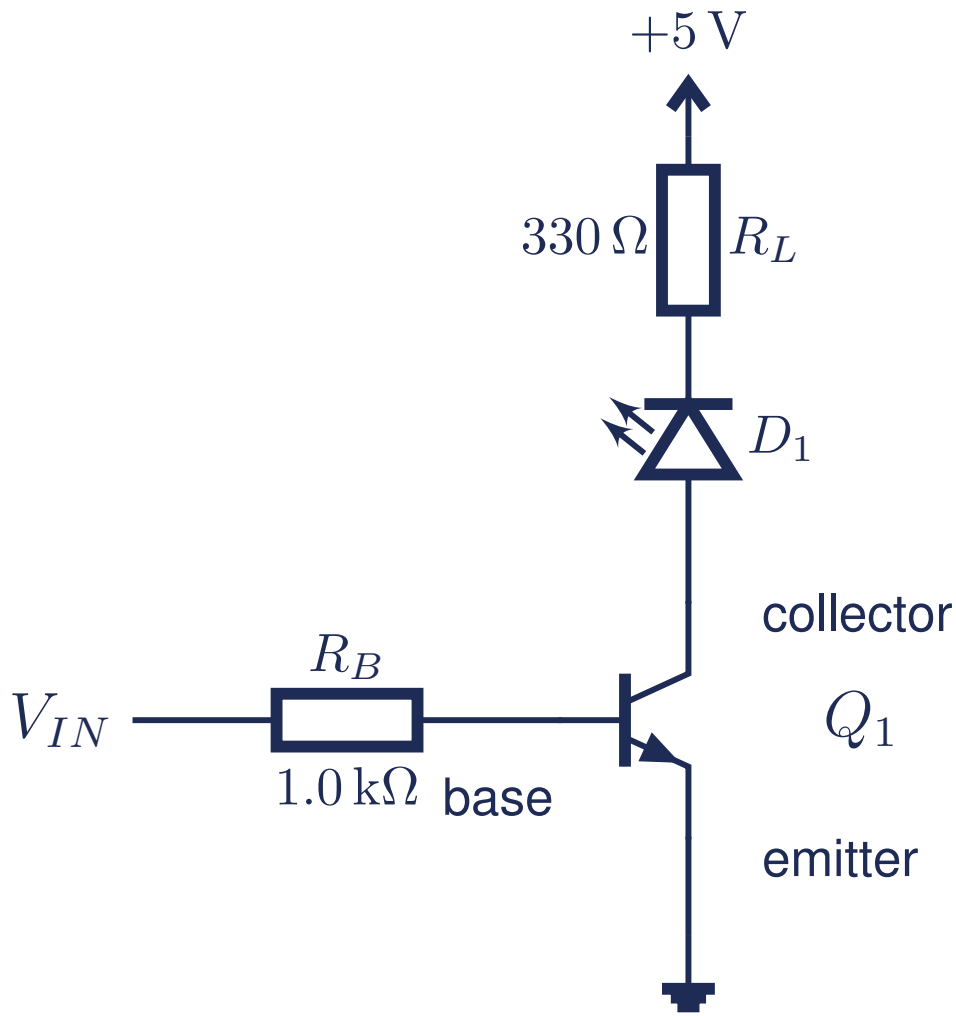
NPN transistor used as a low-side switch



Input current through R_B controls the larger collector current through the LED load.